

DAY 03 · THE AI DEV TOOLCHAIN

One prompt, five hops.

1. Home screen

Every AI product — from ChatGPT to a ten-line Python script — moves a request through the same five stages. Type something and watch it travel.

> Explain APIs like I'm 12

Run ▶

01

User

a person with a goal



02

Prompt

the goal, as text



03

LLM

the model thinks



04

API

the bridge, both ways



05

Application

the answer, delivered

● ● ● trace.log

```
# press Run to trace a request through the pipeline -  
or click any stage below for details
```

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One prompt, five hops.

Every AI product — from ChatGPT to a ten-line Python script — moves a request through the same five stages. Type something and watch it travel.

2. Enter prompt (optional)

> Explain APIs like I'm 12

Run ▶

01

User

a person with a goal



02

Prompt

the goal, as text



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LLM

the model thinks



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DAY 03 · THE AI DEV TOOLCHAIN

One prompt, five hops.

Every AI product — from ChatGPT to a ten-line Python script — moves a request through the same five stages. Type something and watch it travel.

> Generate me 5 topic about economics.|

Run ▶

3. Click 'Run'

01

User

a person with a goal



02

Prompt

the goal, as text



03

LLM

the model thinks



04

API

the bridge, both ways



05

Application

the answer, delivered

● ● ● trace.log

```
# press Run to trace a request through the pipeline –  
or click any stage below for details
```

STAGE DETAIL — CLICK A STAGE ABOVE

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One prompt, five hops.

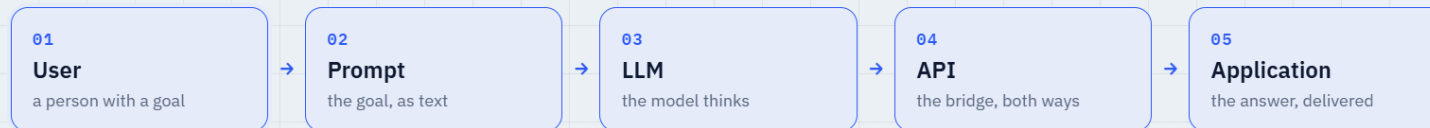
Every AI product — from ChatGPT to a ten-line Python script — moves a request through the same five stages. Type something and watch it travel.

> Generate me 5 topic about economics.

Run ▶



4. The diagram moves and the logger traces the pipeline



● ● ● trace.log

> run

[01 user] intent captured — someone wants an answer

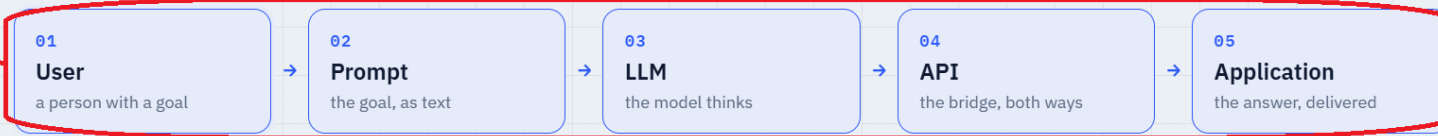
[02 prompt] "Generate me 5 topic about economics." packed into a request

[03 llm] tokens in → tokens out · model is thinking...

[04 api] answer returned as JSON · 200 OK

[05 app] rendered on screen for the user

✓ pipeline complete in 1.77s



● ● ● trace.log

```
> run
[01 user]    intent captured – someone wants an answer
[02 prompt]  "Generate me 5 topic about economics." packed into a request
[03 llm]     tokens in → tokens out · model is thinking...
[04 api]     answer returned as JSON · 200 OK
[05 app]     rendered on screen for the user
✓ pipeline complete in 1.77s
```

5. Choose from the stages above to click and view more details about it

STAGE DETAIL – CLICK A STAGE ABOVE

01 • User

WHAT HAPPENS HERE

Everything starts with a human and a goal — a question, a task, a problem. The user never sees models or servers; they see an input box and expect an answer.

REAL EXAMPLE

Me, opening a chat app and typing: “fix this Python error.”

TOOLS THAT LIVE HERE

None — this step is human. But it defines what every other step has to deliver.

Three tools, one job each

6. A detailed explanation for each of the three main tools:

- API
- CLI
- IDE

INTERFACE

API

A waiter between my code and someone else's service. I never walk into their kitchen — their servers, their model — I just hand a request to the waiter and it comes back with the answer. The menu is fixed: the API defines exactly what I can ask for and how.

```
my code → request → their service  
→ JSON → my code
```

Where it fits: inside the pipeline itself — it's the bridge between my Application and the LLM. Requests go in, answers come back.

TERMINAL

CLI

Talking to the computer with text instead of clicks. One line in the terminal installs a library, runs a script, or pushes code — no hunting through menus. Slower to learn, much faster to use.

```
pip install requests · git push ·  
curl api.model.com
```

Where it fits: around the pipeline, not inside it — it's how I install SDKs, run my script, and test API calls while building.

WORKSHOP

IDE

The workshop where the code gets written. Editor, autocomplete, debugger and a terminal, all in one app — so I'm not juggling ten separate tools while I build.

```
VS Code, PyCharm — mine is VS Code
```

Where it fits: where the Application step is born. The prompt strings and the API calls are all typed here first.

Spot the pipeline in the wild

ChatGPT

A website using AI

A Python script

7. Choose an option to play different scenarios of the same pipeline

User	You, in the browser
Prompt	Your message (+ hidden system instructions)
LLM	A GPT model in OpenAI's data center
API	OpenAI's own API connects the web app to the model
Application	The chat interface at chatgpt.com

It feels like one magic box, but it's the same five hops — OpenAI just built every one of them for you. You only ever touch step 01.